

Q1_ArmstrongClient.java

```
import java.io.*;
import java.net.*;
import java.util.Scanner;

public class Q1_ArmstrongClient {
    public static void main(String[] args) throws Exception {
        Socket s = new Socket("localhost", 5000);
        System.out.print("Enter number: ");
        new DataOutputStream(s.getOutputStream()).writeInt(new Scanner(System.in).nextInt());
        System.out.println(new DataInputStream(s.getInputStream()).readUTF());
        s.close();
    }
}
```

Q1_ArmstrongServer.java

```
import java.io.*;
import java.net.*;

public class Q1_ArmstrongServer {
    public static void main(String[] args) throws Exception {
        ServerSocket ss = new ServerSocket(5000);
        System.out.println("Server waiting...");
        Socket s = ss.accept();
        DataInputStream in = new DataInputStream(s.getInputStream());
        DataOutputStream out = new DataOutputStream(s.getOutputStream());
        int n = in.readInt(), sum = 0, t = n, d = String.valueOf(n).length();
        for (; t > 0; t /= 10) sum += Math.pow(t % 10, d);
        out.writeUTF(n + (sum == n ? " is" : " is not") + " an Armstrong number");
        s.close();
        ss.close();
    }
}
```

Q2_UDPChat.java

```
import java.net.*;
import java.util.Scanner;

// Run "java Q2_UDPChat" for server, "java Q2_UDPChat client" for client
public class Q2_UDPChat {
    static void send(DatagramSocket s, String m, InetAddress ip, int port) throws Exception {
        s.send(new DatagramPacket(m.getBytes(), m.length(), ip, port));
    }

    static DatagramPacket recv(DatagramSocket s) throws Exception {
        DatagramPacket p = new DatagramPacket(new byte[1024], 1024);
        s.receive(p);
        return p;
    }

    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);
        if (args.length == 0) {
            DatagramSocket s = new DatagramSocket(5001);
            System.out.println("Server waiting...");
            while (true) {
                DatagramPacket p = recv(s);
                System.out.println("Client: " + new String(p.getData(), 0, p.getLength()));
                System.out.print("Reply: ");
                send(s, sc.nextLine(), p.getAddress(), p.getPort());
            }
        } else {
            DatagramSocket s = new DatagramSocket();
            while (true) {
                System.out.print("Message: ");
                send(s, sc.nextLine(), InetAddress.getLocalHost(), 5001);
                DatagramPacket p = recv(s);
                System.out.println("Server: " + new String(p.getData(), 0, p.getLength()));
            }
        }
    }
}
```

Q3_URLInfo.java

```
import java.net.URL;

public class Q3_URLInfo {
    public static void main(String[] args) throws Exception {
        URL u = new URL("https://www.example.com:8080/docs/index.html?id=5#top");
        System.out.println("Protocol : " + u.getProtocol());
        System.out.println("Host      : " + u.getHost());
        System.out.println("Port     : " + u.getPort());
        System.out.println("Path    : " + u.getPath());
        System.out.println("File    : " + u.getFile());
        System.out.println("Query   : " + u.getQuery());
        System.out.println("Ref     : " + u.getRef());
    }
}
```

Q4_GUIChat.java

```
import java.awt.*;
import java.net.*;

// Two windows over UDP. Run: "java Q4_GUIChat server" and "java Q4_GUIChat client"
public class Q4_GUIChat {
    public static void main(String[] args) throws Exception {
        boolean server = args.length > 0 && args[0].equals("server");
        int me = server ? 5001 : 5002, peer = server ? 5002 : 5001;
        DatagramSocket s = new DatagramSocket(me);
        Frame f = new Frame(server ? "Server" : "Client");
        TextArea log = new TextArea();
        TextField tf = new TextField();
        Button b = new Button("Send");
        Panel p = new Panel(new BorderLayout());
        p.add(tf);
        p.add(b, BorderLayout.EAST);
        f.add(log);
        f.add(p, BorderLayout.SOUTH);
        f.setSize(300, 300);
        f.setVisible(true);
        f.addWindowListener(new java.awt.event.WindowAdapter() {
            public void windowClosing(java.awt.event.WindowEvent e) { System.exit(0); }
        });
        b.addActionListener(e -> {
            try {
                byte[] d = tf.getText().getBytes();
                s.send(new DatagramPacket(d, d.length, InetAddress.getLocalHost(), peer));
                log.append("Me: " + tf.getText() + "\n");
                tf.setText("");
            } catch (Exception x) { log.append(x + "\n"); }
        });
        while (true) {
            DatagramPacket r = new DatagramPacket(new byte[1024], 1024);
            s.receive(r);
            log.append("Them: " + new String(r.getData(), 0, r.getLength()) + "\n");
        }
    }
}
```